

Homework 6

For Monday, October 5, 2026.

Give exact answers and justify your reasoning. Do not use electronic tools. All functions are real-valued, and angles are measured in radians. You may use the definitions, limit laws and differentiation rules covered through Lecture 7.

1. Let $f(x) = 1/x$, with domain $\mathbb{R} \setminus \{0\}$. Use the definition of the derivative, rather than differentiation rules, to find $f'(a)$ for every $a \neq 0$.
2. Find $p'(x)$ and its domain, where

$$p(x) = 3x^4 - 5x^2 + 7 - \frac{2}{x^3}.$$

3. Suppose f and g are differentiable on an open interval containing 1, and

$$f(1) = 2, \quad f'(1) = -3, \quad g(1) = 4, \quad g'(1) = 5.$$

Find $(fg)'(1)$ and $(f/g)'(1)$.

4. Let

$$r(x) = \frac{x^2 - 4}{x - 2}.$$

Find the domains of r and r' , and find $r'(x)$. Now define $R(x) = r(x)$ for $x \neq 2$ and $R(2) = c$. Determine the value of c that makes R differentiable at 2, and find $R'(2)$.

5. Find and simplify the derivative of

$$F(x) = \frac{\sin x}{1 + \cos x}.$$

State the domains of F and F' .

6. Find the tangent line to $y = 3^x$ where the curve meets the y -axis. Find the x -intercept of this tangent line.

7. Find all points on the graph of

$$h(x) = e^x(x^2 - 3x + 1)$$

at which the tangent is horizontal, and give the corresponding tangent-line equations.

8. Find a polynomial $p(x)$ of degree at most three such that

$$p(0) = 1, \quad p'(0) = 0, \quad p(1) = 0, \quad p'(1) = 0.$$

Show that your polynomial is the only one of degree at most three with these properties.

9. Determine all real numbers a, b for which

$$f(x) = \begin{cases} ax + b, & x < 1, \\ x^2 + 1, & x \geq 1 \end{cases}$$

is differentiable at 1. For those values, give $f'(x)$ on all of $\mathbb{R}$.

10. Let $f(x) = x|x|$. Find $f'(x)$ for $x \neq 0$, then use the definition to determine whether $f'(0)$ exists. Is f' continuous at 0?

11. Let f, g, h be differentiable on an open interval I . Derive a formula for the derivative of fgh using the product rule for two functions. Then apply your formula to find the derivative of $xe^x \sin x$.

12. Evaluate

$$\lim_{x \rightarrow 0} \frac{e^x \cos x - 1}{x}$$

by identifying it as a derivative at a point.

13. Let f, g be defined on an open interval containing a . Suppose that f and fg are differentiable at a , and that $f(a) \neq 0$. Prove that g is differentiable at a , and express $g'(a)$ in terms of $f(a)$, $g(a)$, $f'(a)$ and $(fg)'(a)$. Give a counterexample showing that the hypothesis $f(a) \neq 0$ is necessary.
14. Let f, g be defined on an open interval I containing a . Suppose f is differentiable at a , with $f(a) = f'(a) = 0$, and suppose there is a constant $M > 0$ such that $|g(x)| \leq M$ for every $x \in I$. Prove directly from the definition that fg is differentiable at a and $(fg)'(a) = 0$. No continuity or differentiability of g is assumed.
15. Suppose f is differentiable on $\mathbb{R}$ and is even: $f(-x) = f(x)$ for every $x \in \mathbb{R}$. Use the definition of the derivative to prove that f' is odd, that is, $f'(-x) = -f'(x)$ for every $x \in \mathbb{R}$. What must $f'(0)$ be?